

D203

Fundamentals of Anatomy and Physiology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 34-year-old runner who trains five mornings a week finishes a race on a hot afternoon. Her core temperature rises, and she begins to sweat while blood vessels near her skin widen. She drank two bottles of a sports drink at the halfway point. Which control mechanism is at work?

- A. Voluntary control by the cerebral cortex
- B. Feed-forward control triggered by the sports drink
- C. Negative feedback
- D. Positive feedback

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which of these is an example of positive feedback?

- A. Any response that raises a hormone level is positive feedback.
- B. Oxytocin increases uterine contractions during labor, which trigger more oxytocin release.
- C. Insulin is released after a meal and lowers blood glucose.
- D. Shivering begins when body temperature falls.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

During a trauma assessment, a nurse notes a cut on the patient's forearm, closer to the wrist than to the elbow. The patient, a carpenter, says he was using a new table saw. Which term describes the cut's location relative to the elbow?

- A. Medial
- B. Superficial
- C. Distal
- D. Proximal

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

For a person in anatomical position, which statement uses a directional term correctly?

- A.** The skin is deep to the muscles.
- B.** The knee is proximal to the ankle.
- C.** The nose is lateral to the eyes.
- D.** The wrist is proximal to the elbow.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A student traces a red blood cell that has just picked up oxygen in the lungs. The textbook diagram is color-coded, and the student's lab partner has already labeled the valves. Which heart chamber does this cell enter first?

- A.** Right ventricle, since all blood entering the heart passes through the right side first
- B.** Right atrium
- C.** Left atrium
- D.** Left ventricle

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which vessel carries oxygen-poor blood away from the heart?

- A.** Pulmonary veins
- B.** Superior vena cava
- C.** Aorta
- D.** Pulmonary artery

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Two hours after a large pasta lunch at a company picnic, a healthy adult's blood glucose rises. The picnic ran late and she skipped her usual walk. Which hormone does the pancreas release to bring glucose back toward normal?

- A.** Epinephrine
- B.** Insulin
- C.** Insulin and glucagon in equal amounts, to hold glucose perfectly constant
- D.** Glucagon

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Blood glucose has fallen after an overnight fast. What does glucagon do?

- A.** Signals the liver to break down glycogen and release glucose.
- B.** Signals the kidneys to excrete extra glucose.
- C.** Tells the liver to store glucose as glycogen.
- D.** Moves glucose into muscle cells to lower blood levels.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hiker who has been on the trail since dawn turns a corner and sees a large bear 30 meters away. Her heart pounds, her pupils widen, and her breathing speeds up. Her pack weighs about 15 kg. Which division of the nervous system is dominant?

- A.** Parasympathetic, because the long hike and heavy pack have tired her out
- B.** Somatic nervous system
- C.** Parasympathetic division
- D.** Sympathetic division

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A construction worker spends a summer shift outdoors without drinking enough water, and his blood becomes more concentrated. He works for a firm that just won a large city contract. How do ADH and his urine change?

- A.** ADH rises, the collecting ducts reabsorb more water, and urine becomes small in volume and concentrated.
- B.** ADH falls, and he makes a large amount of dilute urine.
- C.** ADH rises and urine output stops completely.
- D.** ADH rises and the glomerulus stops filtering blood.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a lab on muscle physiology, students review what happens after a motor neuron signal reaches a skeletal muscle fiber. The lab uses a frog leg preparation and runs for three hours. Which event directly allows myosin heads to bind actin?

- A.** Calcium from the sarcoplasmic reticulum binds troponin, moving tropomyosin off the binding sites.
- B.** Sodium binds directly to actin.
- C.** Acetylcholine binds directly to actin.
- D.** ATP binds myosin, causing it to attach to actin.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A patient with healthy lungs is breathing room air. The patient is a retired teacher who swims weekly. In the alveoli, which way does oxygen move, and why?

- A.** From the blood into the alveoli, because blood carries more oxygen.
- B.** In the same direction as carbon dioxide.
- C.** Into the blood by ATP-driven carrier pumps.
- D.** From the alveoli into the capillary blood, because oxygen partial pressure is higher in the alveoli.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A 34-year-old runner ~~who trains five mornings a week~~ finishes a race on a hot afternoon. Her core temperature rises, and she begins to sweat while blood vessels near her skin widen. ~~She drank two bottles of a sports drink at the halfway point.~~ Which control mechanism is at work?

Target word: type of feedback **Decides:** Sweating and vasodilation push temperature back down toward the set point.

A	Adjacent category	Thermoregulation is autonomic, run by the hypothalamus, not a conscious choice.
B	Camouflage	The drink does not decide the mechanism; the response to heat does.
C	Correct	The response reverses the rise in temperature, restoring the set point.
D	Inverted operation	Positive feedback would push temperature even higher.

2. Answer B Concept 1 • transfer

Which of these is an example of positive feedback?

Target word: identify amplification, not correction **Decides:** The output strengthens its own trigger.

A	Absolute language	Feedback type depends on whether the change is reversed or amplified, not on whether a level goes up.
B	Correct	Each round amplifies the next until birth ends the loop.
C	Confusion set	That corrects a rise, so it is negative feedback.
D	Confusion set	Shivering restores temperature; it is negative feedback.

3. Answer C Concept 2

During a trauma assessment, a nurse notes a cut on the patient's forearm, closer to the wrist than to the elbow. ~~The patient, a carpenter, says he was using a new table saw.~~ Which term describes the cut's location relative to the elbow?

Target word: directional term on a limb **Decides:** The wrist end of the forearm is farther from the trunk attachment.

A	Adjacent category	Medial describes the midline axis, not position along a limb.
B	Confusion set	Superficial describes depth, not position along the arm.
C	Correct	Farther from the limb's attachment to the trunk than the elbow.
D	Inverted operation	Proximal means closer to the trunk; the elbow is proximal to the cut.

4. Answer B Concept 2 • transfer

For a person in anatomical position, which statement uses a directional term correctly?

Target word: pick the correct statement **Decides:** The knee is closer to the hip attachment than the ankle.

A	Inverted operation	The skin is superficial to the muscles.
B	Correct	The knee is closer to the trunk attachment than the ankle.
C	Confusion set	The nose is medial to the eyes.
D	Inverted operation	The wrist is distal to the elbow.

5. Answer C Concept 3

A student traces a red blood cell that has just picked up oxygen in the lungs. ~~The textbook diagram is color-coded, and the student's lab partner has already labeled the valves.~~ Which heart chamber does this cell enter first?

Target word: first chamber for oxygenated blood **Decides:** Pulmonary veins return oxygen-rich blood to the left atrium.

A	Absolute language	Blood from the lungs enters the left side directly.
B	Inverted operation	The right atrium receives oxygen-poor blood from the body.
C	Correct	The pulmonary veins empty into the left atrium.
D	Dropped element	Skips the atrium; blood reaches the left ventricle second.

6. Answer D Concept 3 • transfer

Which vessel carries oxygen-poor blood away from the heart?

Target word: vessel by direction and oxygen content **Decides:** Arteries leave the heart; the pulmonary artery goes to the lungs.

A	Inverted operation	They carry oxygen-rich blood toward the heart.
B	Adjacent category	It carries oxygen-poor blood, but toward the heart.
C	Confusion set	The aorta leaves the heart but carries oxygen-rich blood.
D	Correct	It carries oxygen-poor blood from the right ventricle to the lungs.

7. Answer B Concept 4

Two hours after a large pasta lunch ~~at a company picnic~~, a healthy adult's blood glucose rises. ~~The picnic ran late and she skipped her usual walk.~~ Which hormone does the pancreas release to bring glucose back toward normal?

Target word: hormone that lowers glucose **Decides:** Glucose is high after a meal.

A	Adjacent category	An adrenal hormone that raises glucose during stress.
B	Correct	Beta cells release insulin, which moves glucose into cells and storage.
C	Absolute language	The body shifts the balance toward insulin after a meal; glucose is kept in a range, not fixed.
D	Inverted operation	Glucagon raises blood glucose.

8. Answer A Concept 4 • transfer

Blood glucose has fallen after an overnight fast. What does glucagon do?

Target word: glucagon's action **Decides:** Low glucose calls for release from storage.

A	Correct	Glucagon raises blood glucose by releasing liver stores.
B	Adjacent category	Kidneys do not excrete glucose on hormonal signal; glucose falls, not rises, here.
C	Confusion set	Storing glycogen is driven by insulin.
D	Inverted operation	That is insulin's action.

9. Answer D Concept 5

A hiker ~~who has been on the trail since dawn~~ turns a corner and sees a large bear 30 meters away. Her heart pounds, her pupils widen, and her breathing speeds up. ~~Her pack weighs about 15 kg.~~ Which division of the nervous system is dominant?

Target word: which autonomic division **Decides:** A threat triggers fight-or-flight signs.

A	Camouflage	Fatigue details do not decide the response to the bear.
B	Adjacent category	Somatic controls voluntary skeletal muscle, not heart rate or pupils.
C	Inverted operation	Parasympathetic slows the heart and constricts pupils.
D	Correct	Faster heart, dilated pupils, and faster breathing are fight-or-flight effects.

10. Answer A Concept 6

A construction worker spends a summer shift outdoors without drinking enough water, and his blood becomes more concentrated. ~~He works for a firm that just won a large city contract.~~ How do ADH and his urine change?

Target word: ADH direction and urine result **Decides:** Concentrated blood triggers ADH release.

A	Correct	ADH conserves water when the body is dehydrated.
B	Inverted operation	That is the response to overhydration.
C	Absolute language	Output falls but does not stop; waste must still be excreted.
D	Adjacent category	ADH acts on reabsorption in the collecting ducts, not on filtration.

11. Answer A Concept 7

In a lab on muscle physiology, students review what happens after a motor neuron signal reaches a skeletal muscle fiber. ~~The lab uses a frog leg preparation and runs for three hours.~~ Which event directly allows myosin heads to bind actin?

Target word: the step that exposes binding sites **Decides:** Calcium binding troponin shifts tropomyosin.

A	Correct	This exposes actin's binding sites so cross-bridges can form.
B	Confusion set	Sodium drives the action potential; calcium acts on troponin.
C	Dropped element	Skips the steps between the neuromuscular junction and the filaments.
D	Inverted operation	ATP binding causes myosin to release actin.

12. Answer D Concept 8

A patient with healthy lungs is breathing room air. ~~The patient is a retired teacher who swims weekly.~~ In the alveoli, which way does oxygen move, and why?

Target word: direction and driving force **Decides:** Oxygen partial pressure is higher in the alveoli than in the incoming blood.

A	Inverted operation	Blood arriving at the lungs is oxygen-poor.
B	Confusion set	CO ₂ moves the opposite way, from blood to alveoli.
C	Adjacent category	Gas exchange is passive diffusion, not active transport.
D	Correct	Gases diffuse passively down their pressure gradient.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Negative vs. positive feedback	Negative feedback reverses a change to hold a set point; positive feedback amplifies a change until an endpoint is reached.	Calling any response that increases something positive feedback.	Does the response push the variable back toward normal, or further in the same direction?
2	Directional terms	Terms are relative to anatomical position: proximal/distal = closer to or farther from the trunk attachment; medial/lateral = toward or away from the midline; superficial/deep = toward or away from the surface.	Mixing axes, or flipping proximal and distal.	“Relative to” two named landmarks on a limb.
3	Blood flow through the heart	Arteries carry blood away from the heart and veins toward it; oxygen-rich blood returns from the lungs by the pulmonary veins to the left atrium.	Assuming arteries always carry oxygen-rich blood.	A vessel or chamber named in the pulmonary circuit.
4	Blood glucose regulation	Insulin (beta cells) lowers blood glucose by moving it into cells and storage; glucagon (alpha cells) raises it by releasing liver glycogen.	Swapping the roles of insulin and glucagon.	Glucose rising after a meal or falling after a fast.
5	Sympathetic vs. parasympathetic	Sympathetic = fight or flight (faster heart, dilated pupils, airways open); parasympathetic = rest and digest.	Linking stress responses to the parasympathetic side.	A threat or a calm, after-meal setting.
6	ADH and urine concentration	When blood is concentrated, ADH rises and the collecting ducts reabsorb more water, making a small amount of concentrated urine.	Thinking ADH increases urine output.	Dehydration, thirst, or blood concentration.
7	Skeletal muscle contraction	Calcium binds troponin, moving tropomyosin off actin so myosin can bind; ATP is needed for myosin to release and re-cock.	Thinking ATP makes myosin attach.	“What allows cross-bridges to form?”
8	Gas exchange by diffusion	Oxygen and carbon dioxide move passively down their partial pressure gradients: O ₂ from alveoli to blood, CO ₂ from blood to alveoli.	Thinking gases are actively pumped or move the same direction.	Alveoli, capillaries, and the direction of movement.

Course scope (public description): Fundamentals of Anatomy and Physiology provides an overview of the structures and functions of organs and systems of the human body. This course will explore how the parts of the body systems work together to produce movement, transport nutrients, eliminate wastes, protect vital tissues and organs, regulate bodily functions, and support reproduction and growth, through videos, readings, exploratory learning, and practice activities.